



Bus@Belimo

Bus Basics B1



BACnet



Agenda Wednesday



BACnet MS/TP

BACnet IP / Modbus TCP

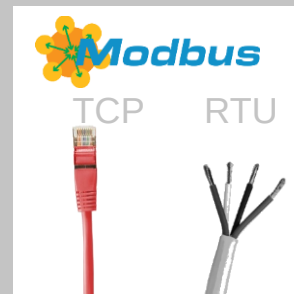
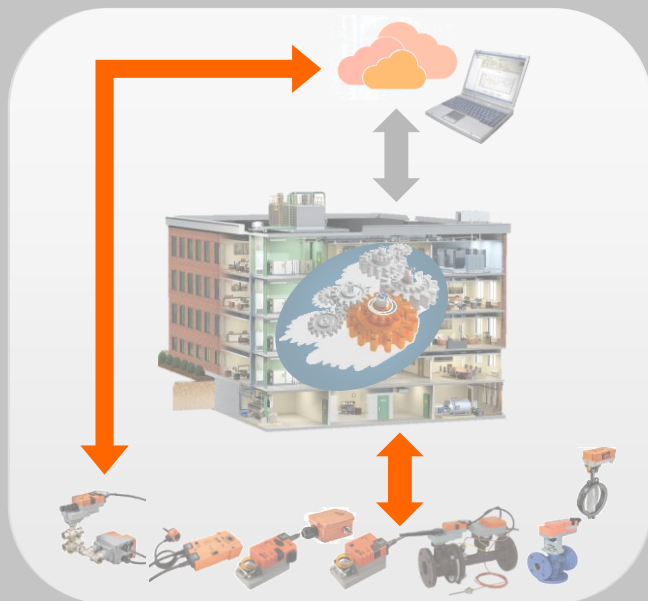
Lunch@Belgusto

MP-Bus

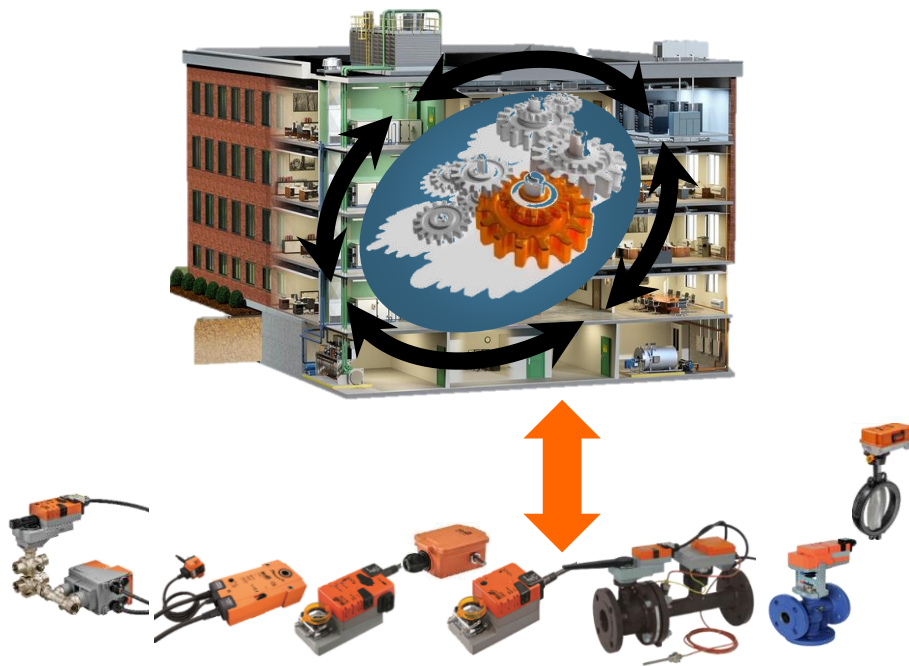
MP-Bus

Dinner together

BACnet@Belimo



Building Technology - History



Many devices

**Many
Manufacturer**

**one Building
one Unit**

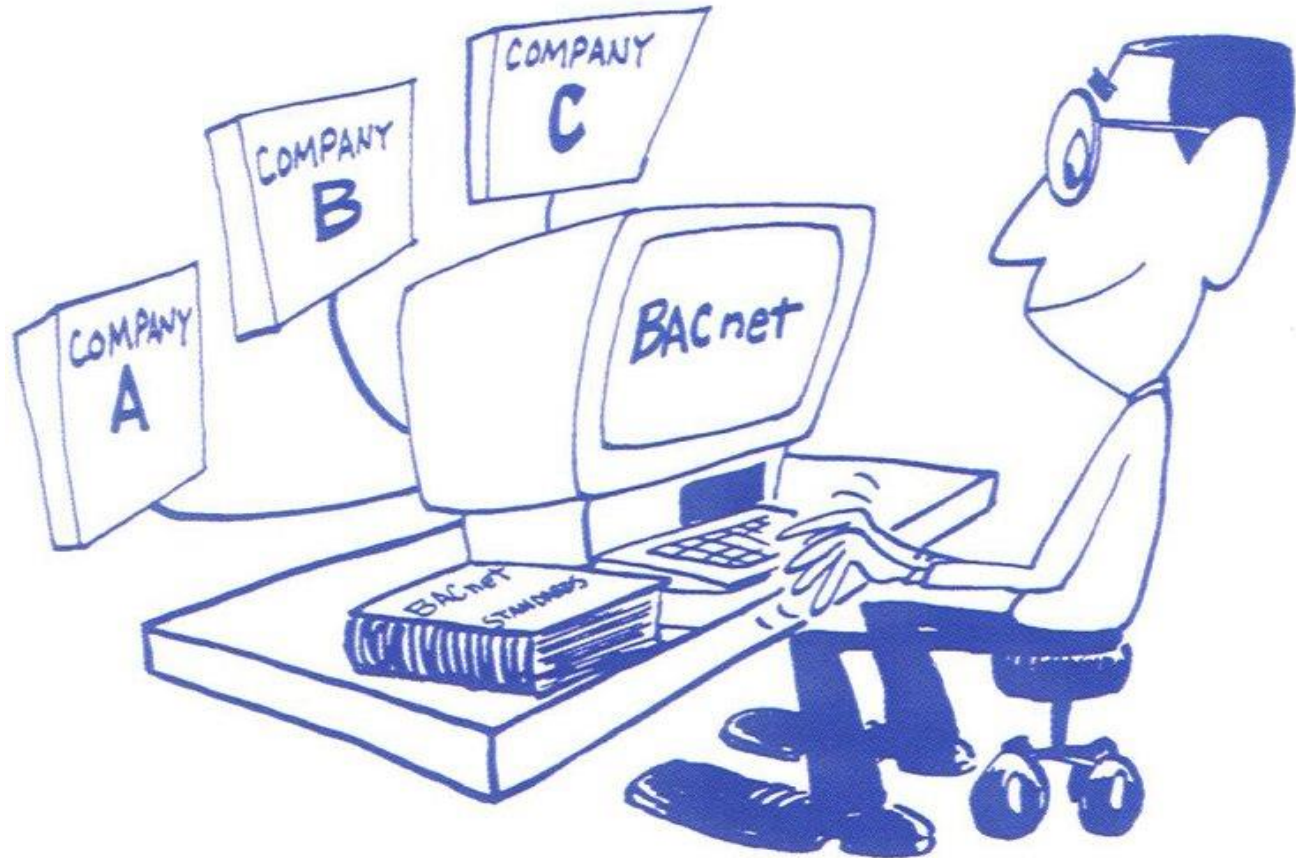
Why do we need BACnet...?

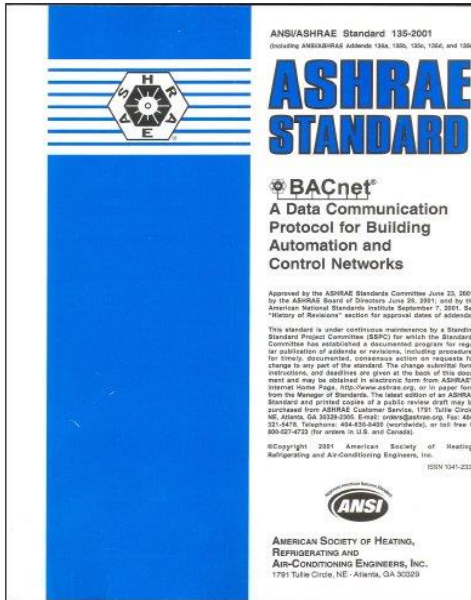
He has a **real problem...**



BACnet - a **Standard** for Building Technology

He's got a solution!





data communication protocol for Building Automation and Control networks

	standard	standard	Vendor ID	Vendor ID
Start	ANSI/ASHRAE 135	ISO 16484-5	423 BELIMO	1000 issued
1987	1995	2003	2010	2017

ASHRAE - official BACnet Web Site: www.bacnet.org

BACnet Interest Group (BIG) - in Europe: www.big-eu.org

BACnet International (BI) - International: www.bacnetinternational.org

BACnet: Objects and Services

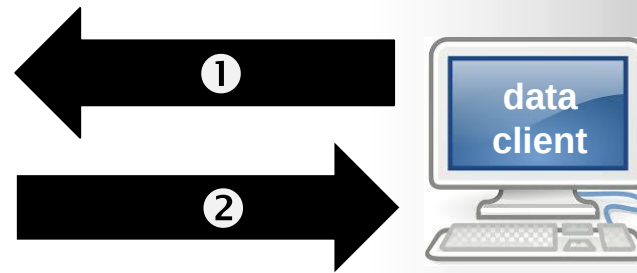
Actuator

BACnet objects:

Rel. setpoint % *Analog Output Object*
Rel. position % *Analog Input Object*
Summary Status *Binary Input Object*
...

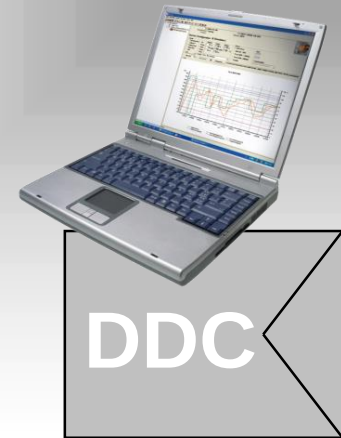
data
server

Automation Station or Management Station



BACnet Services:

- Read data
- Write data
- Notification of value changes (COV)
-



BACnet: PICS and BIBBs

Actuator

BACnet objects:

Rel. setpoint % *Analog Output Object*
Rel. position % *Analog Input Object*
Summary Status *Binary Input Object*
...

BACnet services:

- Read data
- Write data
- Notification of value changes (COV)
-

PICS



Protocol Implementation
Conformance Statement
*Data sheet of the BACnet
communication*

- BACnet Device and Communication Information
- Supported BACnet Objects
- **BIBBs** = Supported BACnet Services

*BIBBs: BACnet Interoperability
Building Blocks*

BACnet: Objects

Actuator

BACnet objects:

Rel. setpoint % *Analog Output Object*

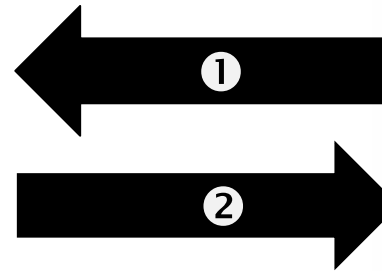
Rel. position % *Analog Input Object*

Summary Status *Binary Input Object*

...

data
server

Automation Station
or
Management Station



DDC

BACnet services:

- Read data
- Write data
- Notification of value changes (COV)
-

BACnet: Objects

Actuator

BACnet Objects:

Rel. setpoint % *Analog Output Object*

Rel. position % *Analog Input Object*

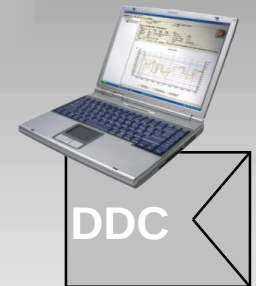
Summary Status *Binary Input Object*

...

data
server

- **BACnet Objects:**

- Outside view of device / actuator
- Every important information is 1 Object
- Each BACnet device has a BACnet Device Object



- **Input / Output names:**

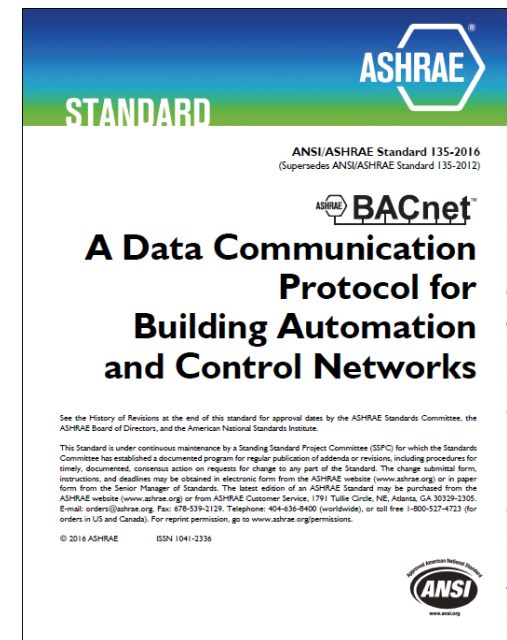
- As if it is already in the DDC / BACnet Client inserted
- Inputs are read only
- Output values can be written
- always from the BACnet Client view

BACnet Objects

BACnet Standard

- Over **30** BACnet Object Types
- There are special objects for the following areas of **Building Technology**:
 - Building automation and HVAC
 - Fire detection and intrusion (Safety & Security)
 - Elevators and other building systems
- Standard Objects** & Proprietary **Objects**

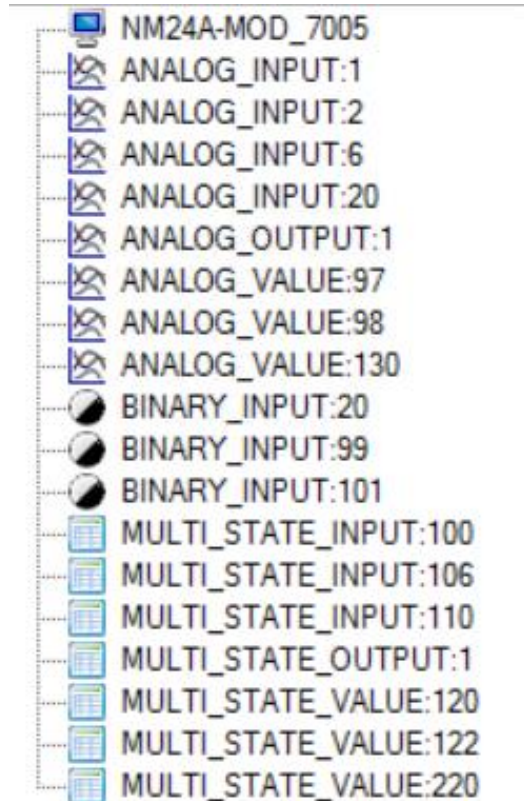
 Binary Input	 Multi-state Input	 File
 Binary Output	 Multi-state Output	 Program
 Binary Value	 Multi-state Value	 Schedule
 Analog Input	 Loop	 Trend Log
 Analog Output	 Calendar	 Group
 Analog Value	 Notification Class	 Event Enrollment
 Averaging	 Command	 Device
 LifeSafetyZone	 LifeSafetyPoint	



BACnet Objects

Example: NM24A MOD

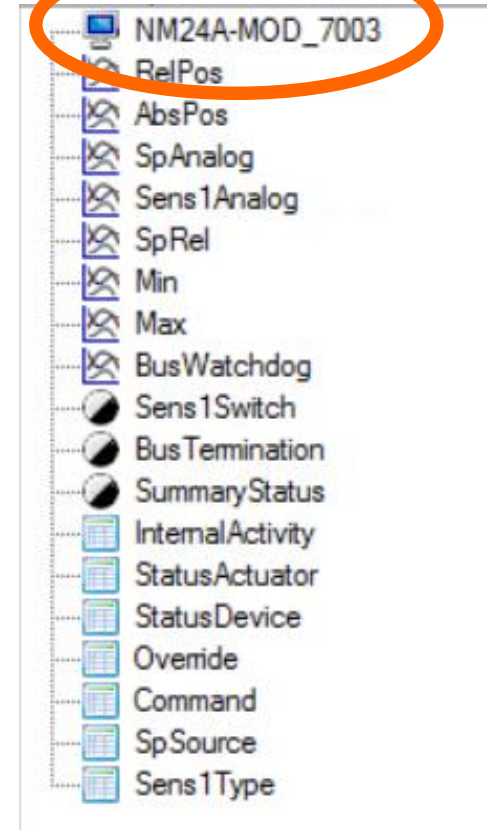
Device Object



object type



Read Property:
Object Name



BACnet Device Object - Device Identity



- BACnet system-wide unique name and **instance number = BACnet address**
 - Instance Number: 22 Bit 2^{22} Devices in the system
- Vendor ID, Software Version, Device Description, Location Text, List of Properties

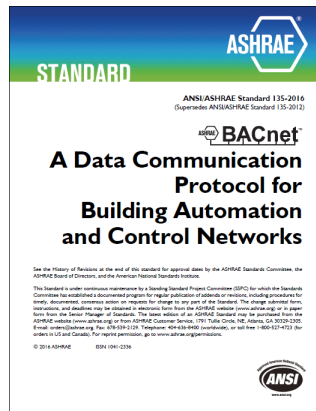


Table 12-13. Properties of the Device Object Type

Property Identifier	Property Datatype	Conformance Code
Object_Identifier	BACnetObjectIdentifier	R
Object_Name	CharacterString	R
Object_Type	BACnetObjectType	R
System_Status	BACnetDeviceStatus	R
Vendor_Name	CharacterString	R
Vendor_Identifier	Unsigned16	R
Model_Name	CharacterString	R
Firmware_Revision	CharacterString	R
Application_Software_Version	CharacterString	R
Location_Text	CharacterString	O
Device_Description	CharacterString	O
Protocol_List	CharacterString	R
Protocol_List	CharacterString	R
Protocol_List	CharacterString	R
Protocol_List	CharacterString	R
Protocol_List	CharacterString	R
Object_Identifier	CharacterString	R
Structure	CharacterString	O
Max_Segments_Supported	BACnetSegmentation	R
Max_Segments_Accepted	Unsigned16	O

Each object has a set of properties:
Required / **Optional** / **Proprietary**

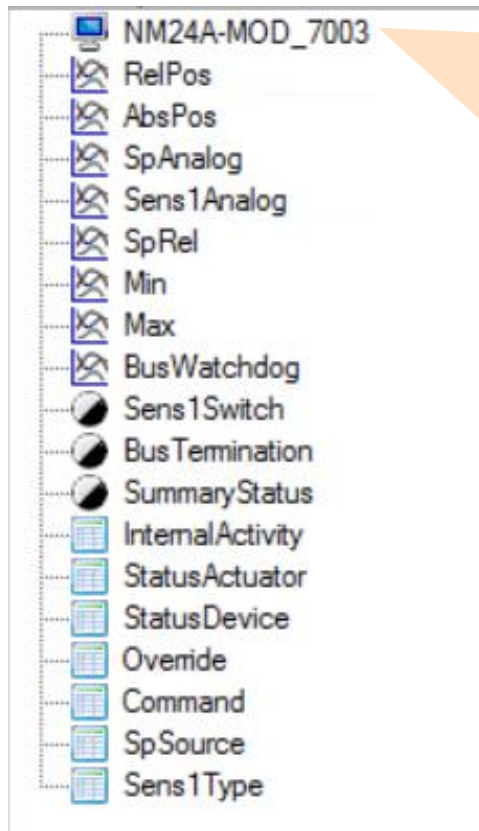
BACnet Instance Number = Address

➔ *BACnet Instance Number & Device Name must be unique in the entire BACnet network!*



BACnet Objects and Properties

Example: NM24A MOD



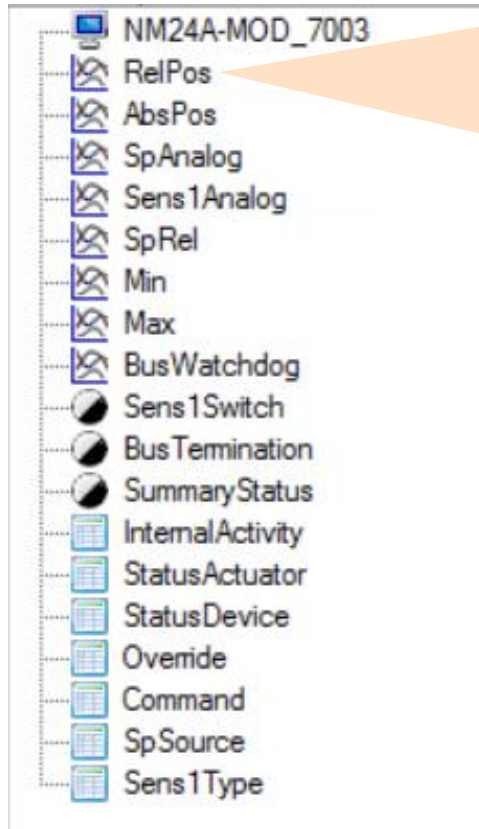
Device Object: NM24A-MOD_7003

BacnetProperty	
Active Cov Subscriptions	
Apdu Timeout	3000
Application Software Version	03.03.0006
Database Revision	4
Description	
Device Address Binding	
Firmware Revision	08.05.0000
Location	
Max Apdu Length Accepted	480
Max Info Frames	10
Max Master	127
Model Name	BACMFT for NM24A-MOD, 01736-30137-142-126
Number Of Apdu Retries	3
> Object Identifier	OBJECT_DEVICE:7003
> Object List	Object[]-Array
Object Name	NM24A-MOD_7003
Object Type	8 : Object Device
Profile Name	No profile set
Protocol Object Types Supported	1111000010000110000100000000000000000000
Protocol Revision	12
Protocol Services Supported	0000010000001011110000000000000001100000
Protocol Version	1
Segmentation Supported	3 : None
System Status	0 : Operational
Vendor Identifier	423
Vendor Name	BELIMO Automation AG

BACnet Objects and Properties

Example: NM24A MOD

Analog Input Object: RelPos



Bacnet Property	
Cov Increment	1
Description	Relative Position in %
Event State	0 : Normal
Object Identifier	OBJECT_ANALOG_INPUT:1
Object Name	RelPos
Object Type	0 : Object Analog Input
Out Of Service	False
Present Value	50.73
Status Flags	0000
Units	98 : Percent

Example Relinquish Default

VAV Compact: Bus Watchdog

from PICs

AV[130]	Timeout for Bus Watchdog in s 0 = watchdog deactivated. If the Present_Value is not ZERO, the implementation tracks write procedures to Present_Value of AO[1] and MO[1]. If the Present_Value of AO[1] or MO[1] is written, the timer is reset. Upon timeout the Priority_Array of the AO[1] is cleared and the Relinquish_Default becomes valid.	0...3600
---------	---	----------

AO[1]	Relative Setpoint in %	0...100
-------	------------------------	---------

Relinquish Default = Default / start value of the Present Value

BACnet: Services

Actuator

BACnet objects:

Rel. setpoint % *Analog Output Object*

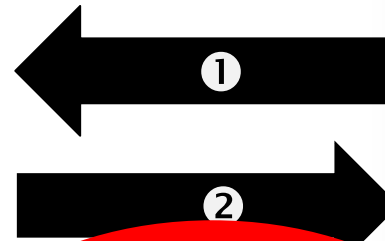
Rel. position % *Analog Input Object*

Summary Status *Binary Input Object*

...

data
server

Automation Station or Management Station

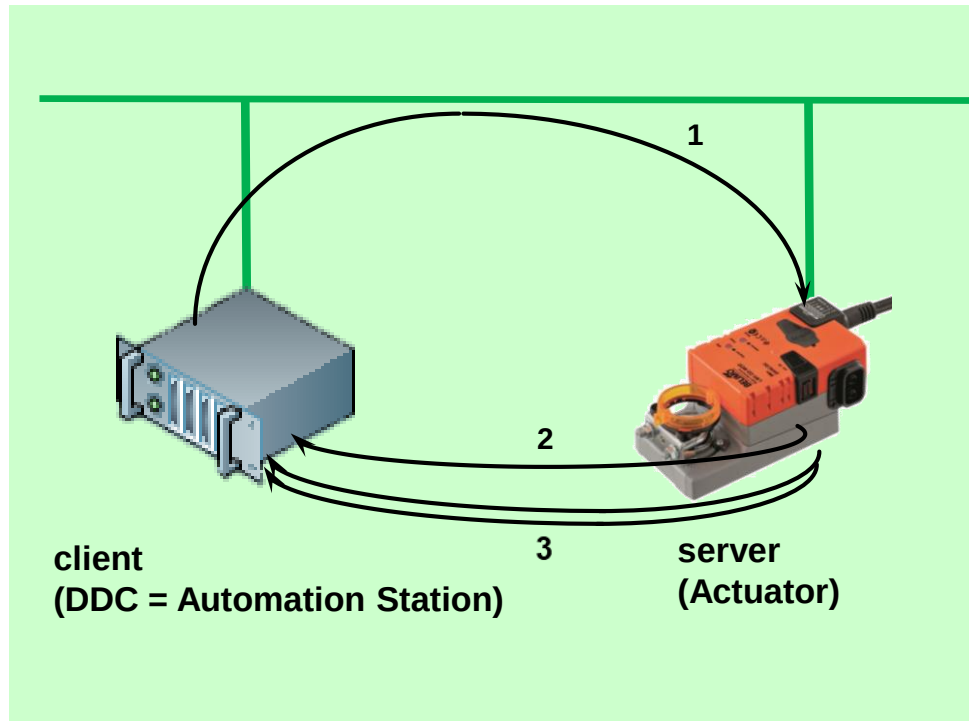


BACnet services:

- Read data
- Write data
- Notification of value changes (COV)
-

BACnet COV better than polling

Example: Change of Value (COV) Service



1. Client (DDC) subscribes to a value (datapoint) of the server (Actuator)
2. Server (Actuator) acknowledges this and sends the current value to the client (DDC).
3. After changing the value, the server (Actuator) automatically sends the new value to the client (DDC).

BACnet Communication

Transport data between clients and server (devices)

Actuator

BACnet objects:

Rel. setpoint % *Analog Output Object*

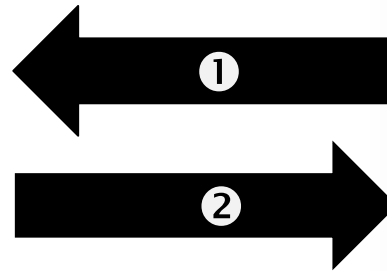
Rel. position % *Analog Input Object*

Summary Status *Binary Input Object*

...

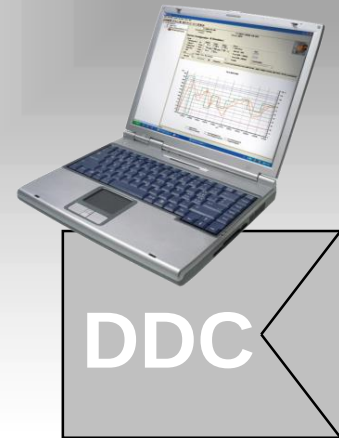
data
server

**Automation Station
or
Management Station**



BACnet services:

- Read data
- Write data
- Notification of value changes (COV)
-



BACnet Protocol Language: Transport Options



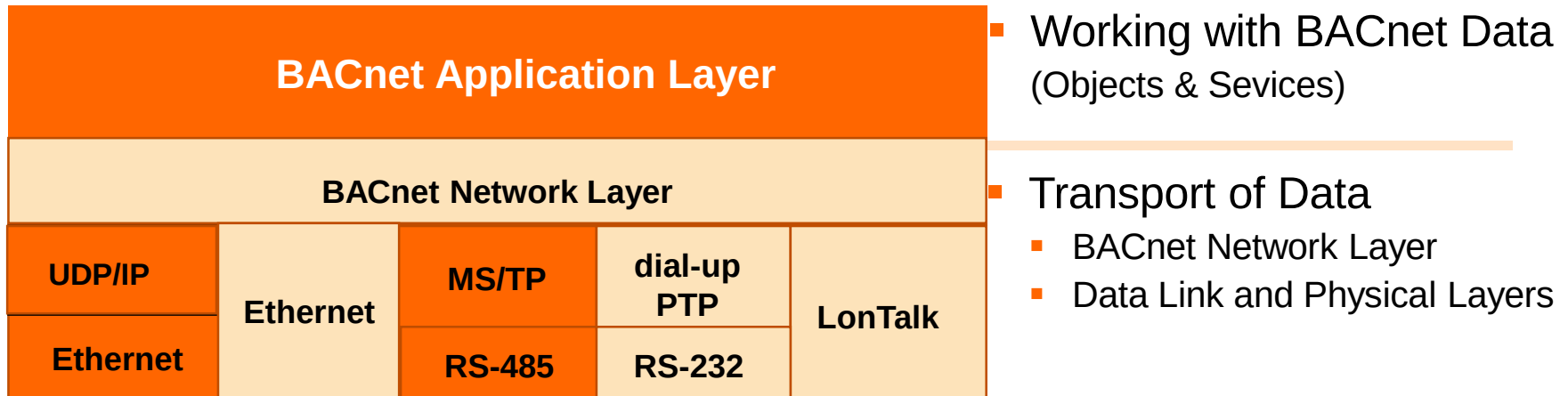
Transport options:

BACnet IP
BACnet MS/TP
BACnet PTP
BACnet Ethernet
BACnet ARCNET
BACnet over ZigBee
BACnet LonTalk
BACnet WS

....



BACnet Communication Layers

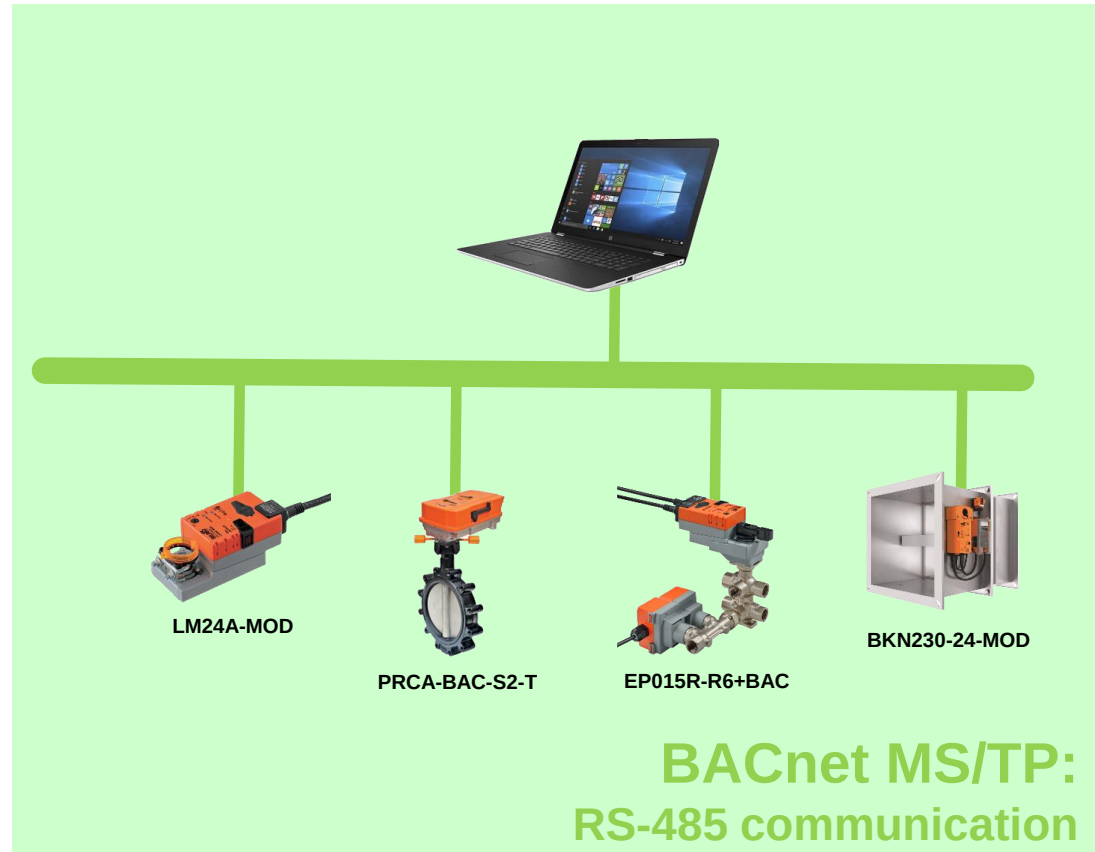


Transport Options @ Belimo

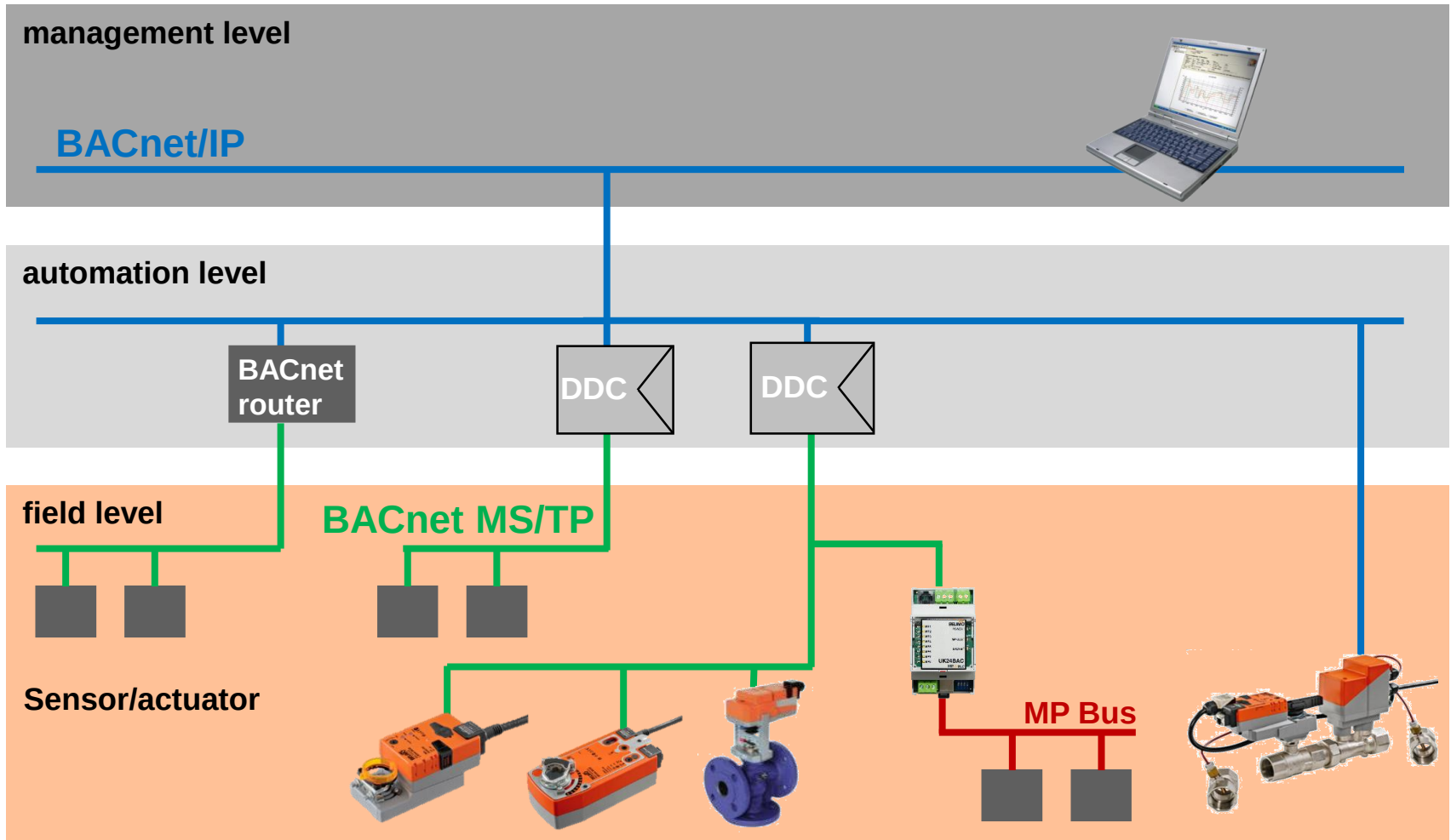
BACnet IP UDP - ISO-8802-2 - Ethernet

BACnet MS/TP - RS485

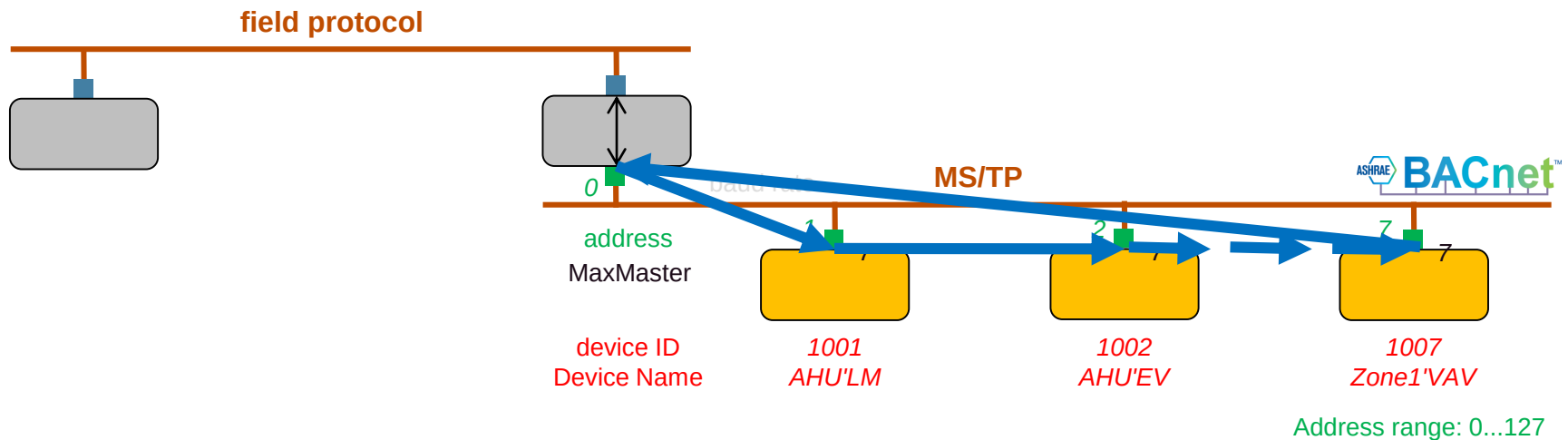
BACnet IP and BACnet MS/TP



BACnet IP and BACnet MS/TP

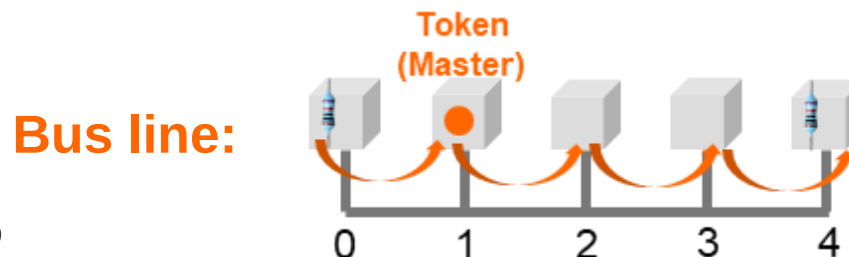


BACnet MS/TP Parameters & Communication



MS/TP: RS-485 wiring

- Industry standard **EIA-485** for asynchronous data transmission
- Bus / line technology (Daisy Chain):
 - Stub lines generally not allowed
 - ⇒ **Therefore there are only short Actuator cable lengths for RS-485 communication such as Modbus RTU and BACnet MS/TP.**
- Longer cables must be terminated with a **terminating resistor** on both sides!
 - Install terminating resistor physically
 - if supported Switch on terminating resistor by software (e.g. with ZTH / App)
- **MS/TP:**
 - Master-Slave - Token Passing
 - We **only** use **masters**, all masters use the token for communication.



Translation: **MP** ⇔ **BACnet MS/TP Gateway**



Language translation / Protocol translation

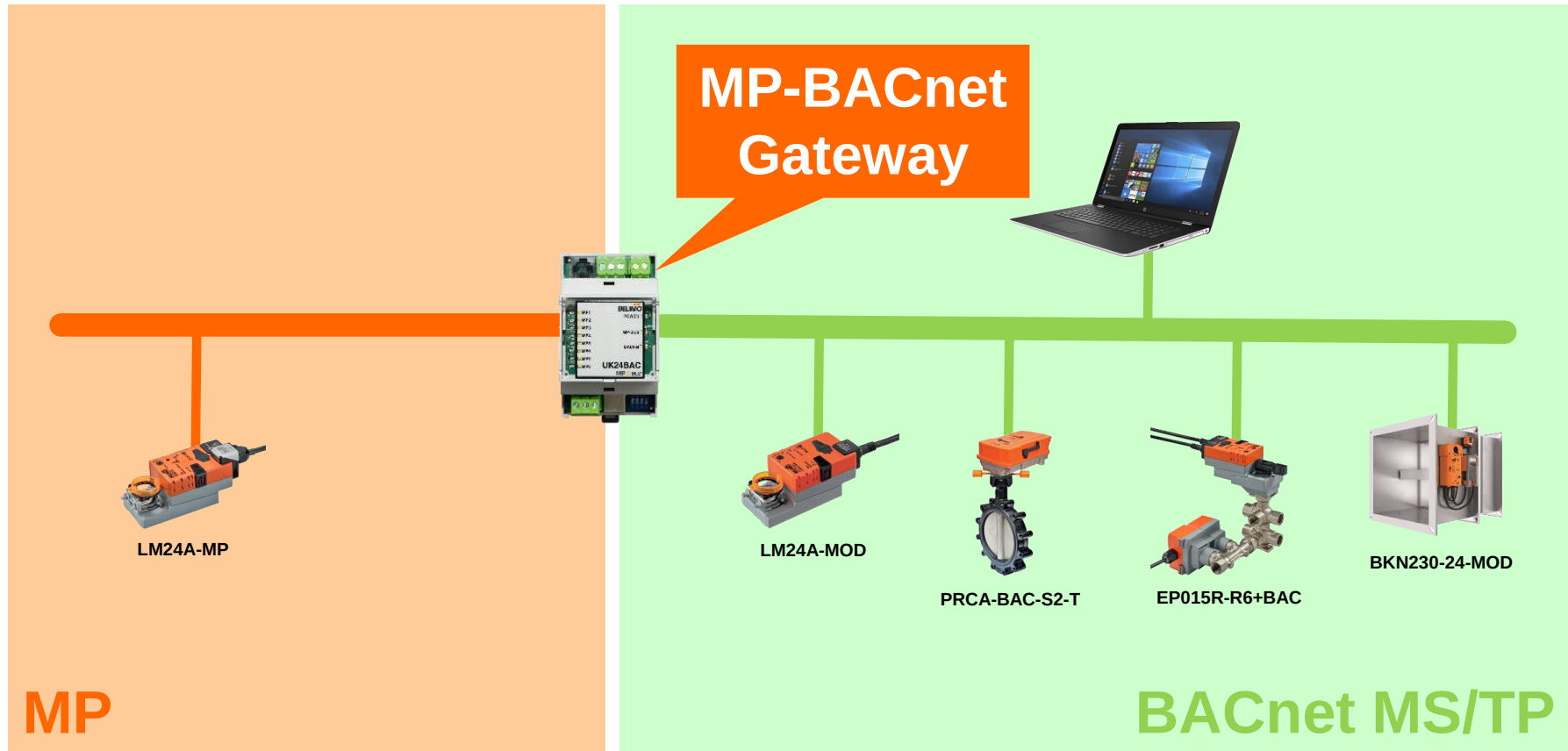


Gateway

UK24BAC:

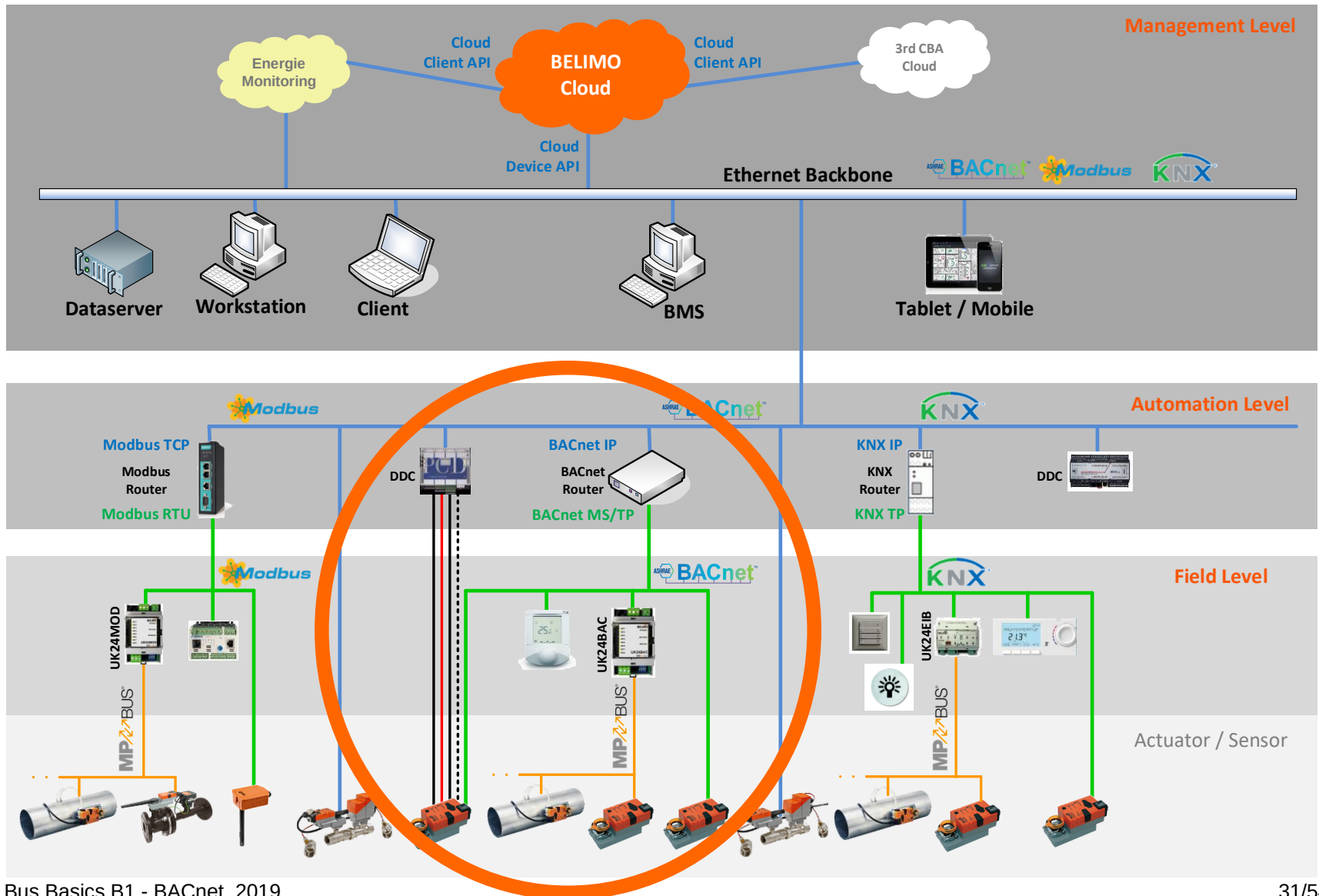
MP ⇔ BACnet MS/TP

Example Gateway



Attention: Not all actuator / sensors are supported by the gateways!

BACnet @Belimo



BACnet EDE File



How can BACnet objects of a device or system be exported to a file?

Engineering Data Exchange Template for BACnet Systems

- Excel file (CSV)
- Export from manufacturer specific engineering tool or browse and export with tool like yabe

BACnet@Belimo Product Portfolio



				
L...24A-MOD N...24A-MOD S...24A-MOD	SF24A-MOD	VAV-Compact	BKN230-24MOD	CQ24A-BAC
				
PRCA...-BAC	NV24A MOD	2-way EPIV	6-way EPIV	Energy Valve
				
Sensor Products	IoT Product Range		UK24BAC	



BACnet@Belimo Product Portfolio

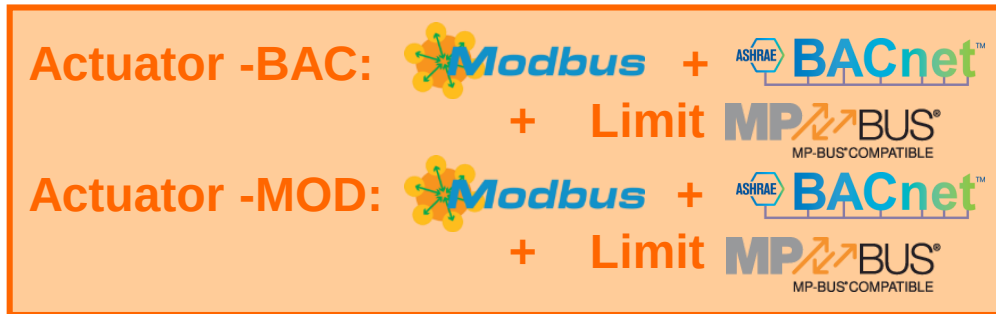
BACnet IP and BACnet MS/TP



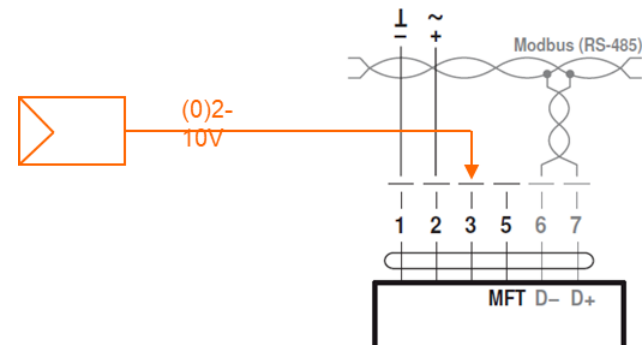
L...24A-MOD N...24A-MOD S...24A-MOD	SF24A-MOD	VAV-Compact BACnet MS/TP	BKN230-24MOD	CQ24A-BAC
PRCA...-BAC	NV24A MOD BACnet MS/TP	2-way EPIV	6-way EPIV	Energy Valve BACnet IP
Sensor Products MS/TP	IoT Product Range BACnet IP			UK24BAC MS/TP



Portfolio Actuator Modbus / BACnet



- Backwards compatible
- Hybrid mode for Modbus and BACnet
 - Control Analog
 - Monitoring and override control Bus
- MP Bus supports with Limitation:
 - no COM LED for MP
 - no hybrid mode



Bus@Belimo Portfolio in Detail

Bus Brochure and Device Selection Tool



www.belimo.eu/busdevice

Device selection tool

Filter bus devices

Direct integration

- ☐ MP-Bus
- ☒ **Modbus**
 - ☒ RTU
 - ☐ TCP
- ☐ BACnet
- ☐ LON
- ☐ KNX
- ☐ Cloud

Sensor connection

- ☐ Active (0-10 V)
- ☐ Active (0-32 V)
- ☐ Passive (e.g. PT1000)
- ☐ Switching contact

Type

- ☒ Actuator
- ☐ Field device
- ☐ Sensor

Reset

MP-Bus with gateways

- ☐ via UK24MOD
- ☐ via UK24BAC
- ☐ via UK24LON
- ☐ via UK24EIB

Application

- ☐ Air
- ☐ Safety
- ☐ VAV
- ☐ Water

Fail-safe function

- ☐ with
- ☐ without

66 suitable devices

AVK24A-MOD

Globe v
AC/DC
Commu
MP-Bus
Running
Running
IP54, M

CQ24A-BAC

Rotary a
AC/DC
Commu
Running
IP40, M
Connec

EP015R+MOD

Electr. 2
Internal
AC/DC
Commu
MP-Bus
Adjusta
IP54, M
locked

EP015R-R6+BAC

Electr. 6
Internal
AC/DC
Commu

37/54

PICS Exercise

1. Is this BACnet Actuator officially tested to be according to the BACnet Standard? _____
2. Which kind of BACnet Communication are supported by this Device: _____
3. What is the default Baud rate? _____
4. How many COVs are supported? _____
5. How many and which Access Types are available for BACnet Objects?

PICS Exercise



1. Is this BACnet Actuator officially tested to be according to the BACnet Standard?

Conformation

Listed by BTL

2. Which kind of BACnet Communication are supported by this Device:

Segmentation capability

No

Data Link Layer Options

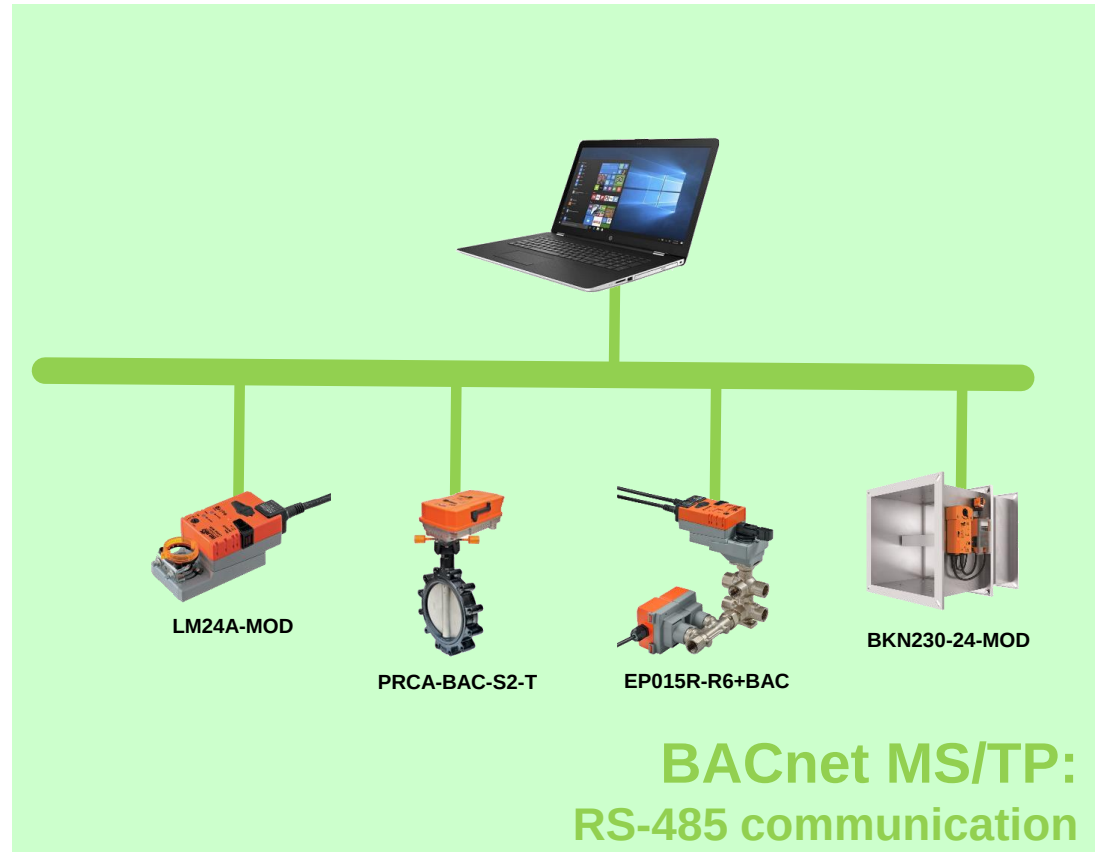
MS/TP master

3. What is the default Baud rate? 38'400
4. How many COVs are supported? 6 active COV Subscriptions
5. How many and which Access Types are available for BACnet Objects?

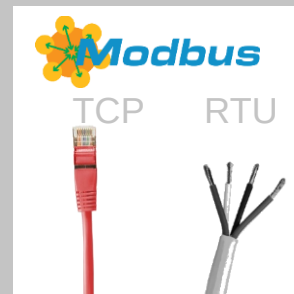
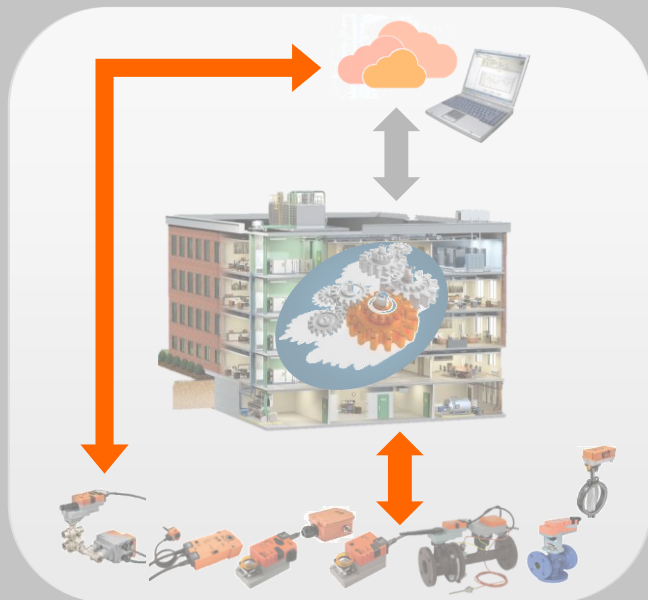
Access: R = Read, W = Write, C = Commandable with priority array

BACnet MS/TP

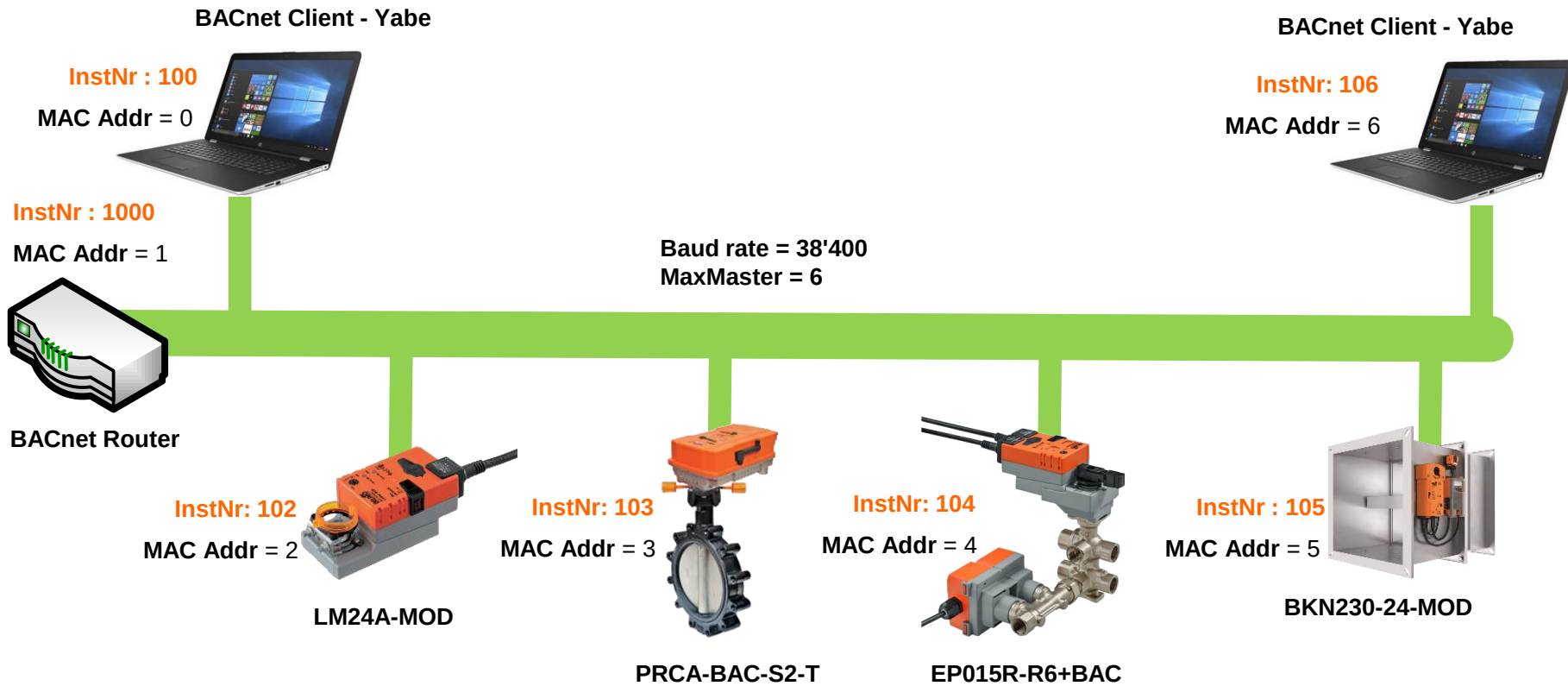
RS-485 Serial Communication



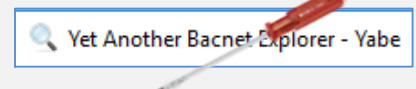
BACnet@Belimo



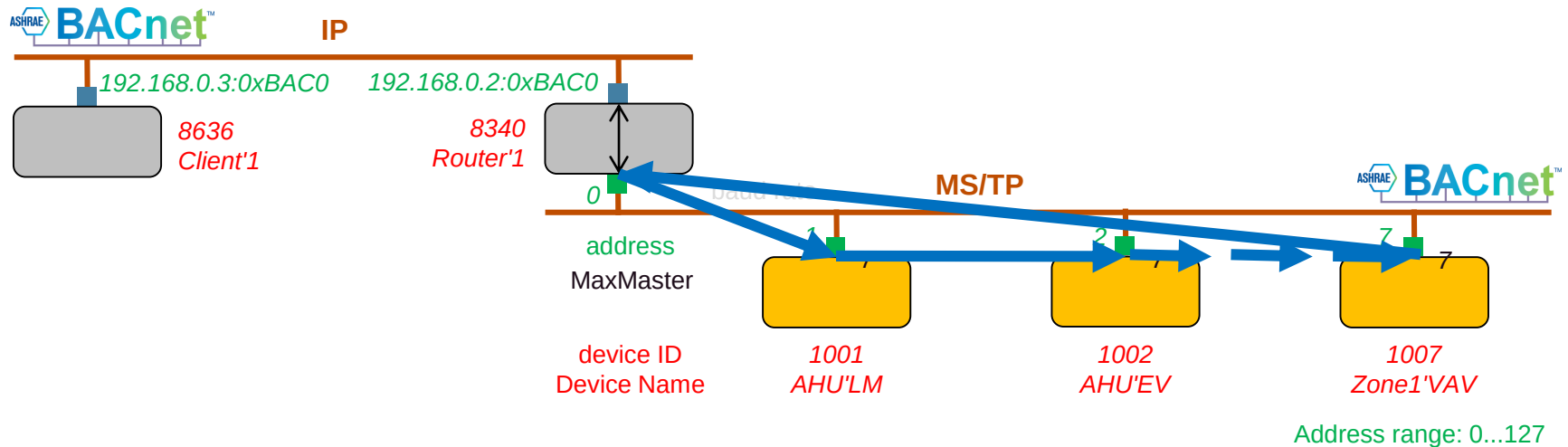
BACnet MS/TP Addressing



Parameterisation

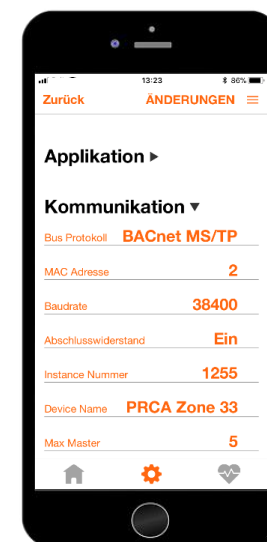
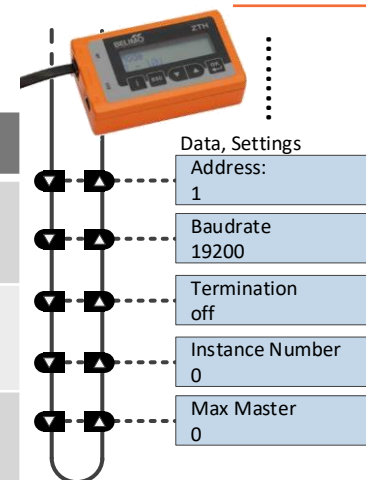


BACnet MS/TP Parameters & Communication



BACnet MS/TP Configuration for Belimo Devices

Device Name	Site4'Floor2 'Zone3'VAV1	BACnet Internetwork-wide unique Text for the Device Identification
Instance Number	1 - 4194302	BACnet Internetwork-wide unique Number for the Device Identification
MAC Address	0-127	MS/TP MAC Address, unique in each MS/TP Line
Baud rate	9'600 19'200 38'400 76'800 115'200	Baud rate of the RS485 Network . Insert the Terminating Resistor at the first and last device of the line. Supported baud rates of the respective device can be found in the PICS.
Max Master	0-127	Last MS/TP device address of the line. Optimizes the communication. Set to highest MAC address of the MS/TP line.

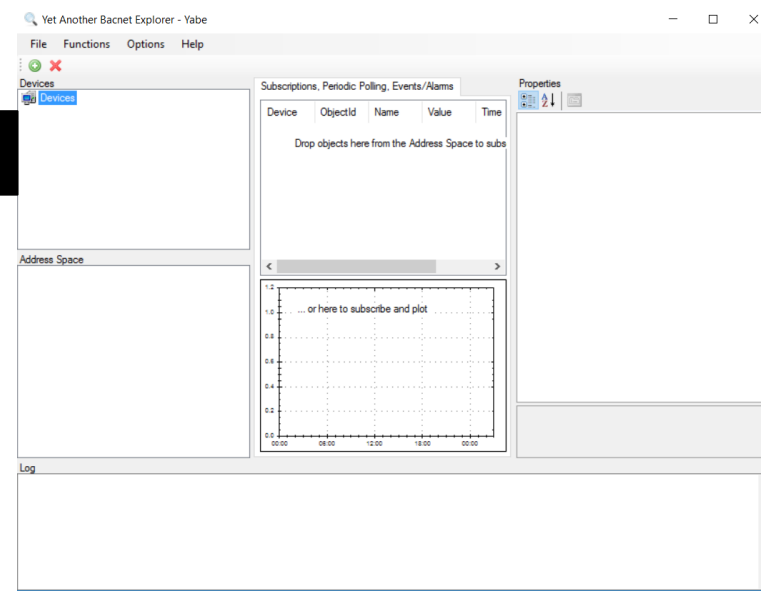
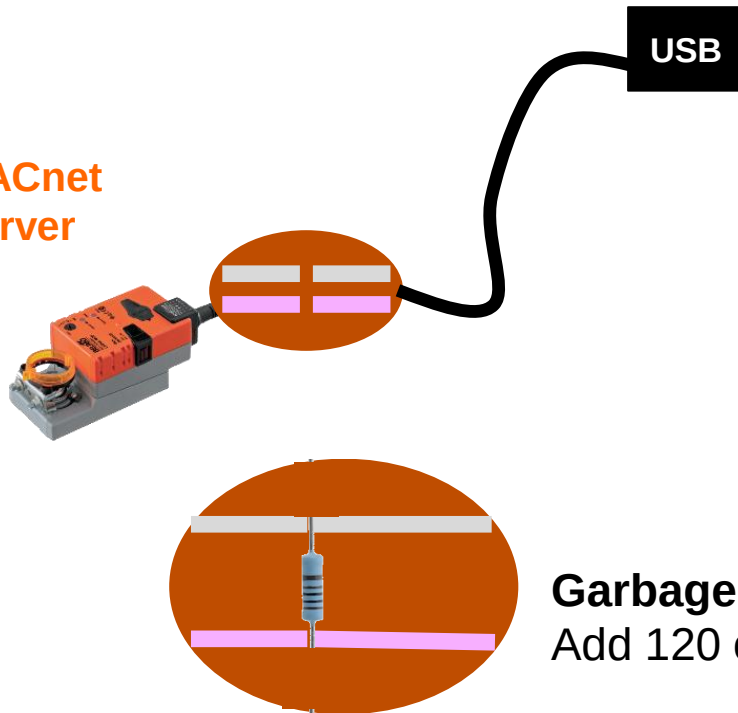


- ➔ Check max. number of Devices per MS/TP Line (typically 32 Devices)
- ➔ BACnet Instance Number and Device Name must be unique in the entire BACnet Internetwork!

Workshop Tool Yabe and cabling

BACnet Client

BACnet
server



Garbage in the Yabe Tool:
Add 120 ohm resistance between pink&grey

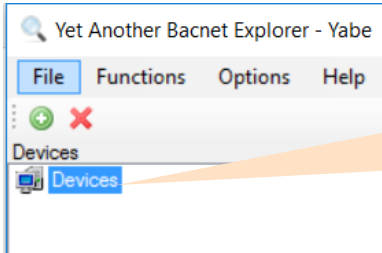
BACnet Client - Example YABE

The screenshot displays a BACnet client interface with three main sections:

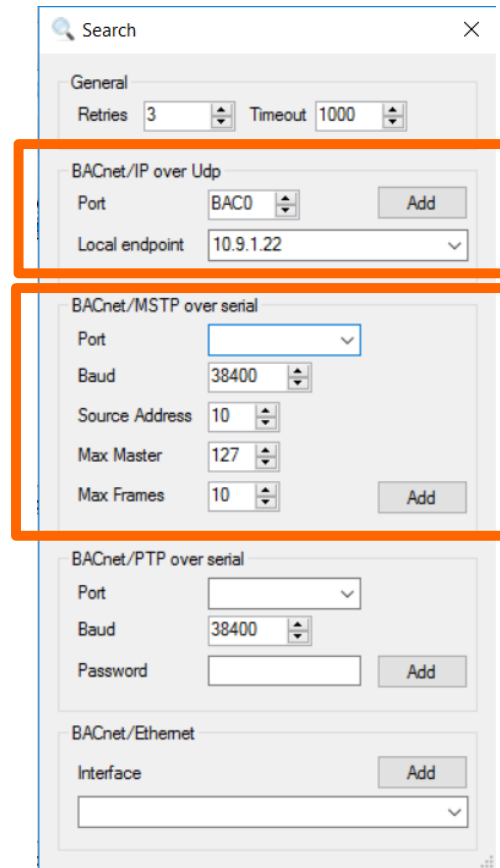
- Devices:** A tree view showing a hierarchy of devices. Under 'Udp:47809', there are several devices including 'Device 748 - 1 via 10.20.25.148:47809', 'Device 749 - 1 via 10.20.25.149:47809', 'Device 1000 - 10.20.27.198:47809', and a group of objects under 'Device 1000' such as 'LHV-D3-MOD200 AU_1251 [1251]', 'LM24A-MOD_1252 [1252]', 'AVK24AX-MOD_12007 [1253]', 'PR-8 [1254]', 'BKN230-24-MOD_1255 [1255]', and 'BACDOC [1258]'. Below these are 'Device 25 - 12 via 10.20.25.148:47809', 'Device 26 - 13 via 10.20.25.148:47809', and 'Device 27 - 14 via 10.20.25.148:47809'.
- Address Space:** A list of objects for the selected device 'LM24A-MOD_1252'. The objects include 'RelPos', 'AbsPos', 'SpAnalog', 'Sens1Analog', 'SpRel' (highlighted), 'Min', 'Max', 'BusWatchdog', 'Sens1Switch', 'BusTermination', 'SummaryStatus', 'Override', 'Command', 'SpSource', and 'Sens1Type'.
- Sub Properties:** A table showing properties for the selected object 'BacnetProperty'. The table has two columns: the property name and its value.

Sub Properties	
BacnetProperty	
Cov Increment	1
Description	Relative Setpoint in %
Event State	0 : Normal
Object Identifier	OBJECT_ANALOG_OUTPUT:1
Object Name	SpRel
Object Type	1 : Object Analog Output
Out Of Service	False
Present Value	22
Priority Array	Object[]-Array
Relinquish Default	0
Status Flags	0000
Units	98 : Percent

Below the table, the 'Cov Increment' is shown as 'BACNET_APPLICATION_TAG_REAL'.



**Right click:
add device**



BACnet IP

BACnet MS/TP

Main Advantages of BACnet



- **More specific Information** available, helpful during project engineering, commissioning and maintenance
Units of the Value, Project specific Descriptions for all objects, Device Location names, Actuator names
- **Automatic Notification (COV)** in case of any value changes or needed maintenance
For example: actuator position changes, occurred errors like “actuator cannot move”
- **Standard Integration** via well known BACnet Objects
Standard Integration to all kind of Controllers and Management stations possible with low effort

Differences

Comparison between BACnet and Modbus protocol

The screenshot displays a BACnet configuration software interface. On the left, a 'Devices' tree lists various devices, including 'LM24A-MOD_1252'. Below it, the 'Address Space' for 'LM24A-MOD_1252' is shown, listing objects like 'RelPos', 'AbsPos', 'SpAnalog', 'Sens1Analog', 'SpRel', 'Min', 'Max', 'BusWatchdog', 'Sens1Switch', 'BusTermination', 'SummaryStatus', 'Override', 'Command', 'SpSource', and 'Sens1Type'. The 'SpRel' object is selected. On the right, the 'Properties' window for 'SpRel' is displayed, showing the following data:

Bacnet Property	
Cov Increment	1
Description	Relative Setpoint in %
Event State	0 : Normal
Object Identifier	OBJECT_ANALOG_OUTPUT:1
Object Name	SpRel
Object Type	1 : Object Analog Output
Out Of Service	False
Present Value	22
Priority Array	Object[]-Array
Relinquish Default	0
Status Flags	0000
Units	98 : Percent

Below the properties table, the 'Cov Increment' is set to 'BACNET_APPLICATION_TAG_REAL'. To the right of the properties window, a diagram shows two twisted-pair cables connected by a BACnet MS/TP (Master/Slave Transmittal Protocol) link, with the ASHRAE BACnet logo above it.

Differencies

BACnet compared to Modbus protocol

- **BACnet** offers more **features** like
 - units of the values 100% ⇔ 10000
 22 l/min ⇔ 22000
 - Text, Floatingpoint values, etc.
 20.31 °C ⇔ 2031
- **BACnet** communication results in more **network load**
 - Exchange of more information using mechanism with more overhead
- **Modbus** uses a very **compact form without any overhead**
 - Allows more efficient and faster exchange of information

Any questions?





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